Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LEARNER GUIDE

For

Fundamentals of ISO 14001 Environmental Management System (EMS)

TGS Ref No: TGS-2023037592

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 8.0

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
7.0	Legacy release (date not stated)	Legacy trainer-slide baseline.	Dr Alfred Ang
8.0	20 August 2026	Rebuilt for ISO 14001:2026 with eight evidence-led EMS activities and aligned assessment preparation.	Dr Alfred Ang

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

TABLE OF CONTENTS

How to Use This Guide **4**
Course Learning Outcomes **4**
TSC Knowledge and Abilities **4**
Connected Case Organisation **5**
1. EMS Foundations and ISO 14001 ..... **6**
Core working terms 6
PDCA and intended outcomes 6
2. Context, Leadership and Planning — Clauses 4 to 6 ..... **8**
A defensible aspect-significance method 8
Compliance obligations workflow 8
3. Support and Operation — Clauses 7 and 8 ..... **10**
Life-cycle perspective without requiring a full LCA 10
Emergency preparedness and response 10
4. Performance Evaluation — Clause 9 ..... **11**
Writing a defensible audit finding 11
5. Improvement and ISO 14001:2026 Transition — Clause 10 ..... **12**
A controlled 2026 transition 12
6. Detailed Activity Procedures **13**
Activity A1 — Build the Environmental Baseline and Aspect Inventory (A1) 13
Activity A2 — Assess Significance, Risks and Opportunities (A2) 15
Activity A3 — Build the Compliance Obligations Register (A4) 17
Activity A4 — Set Environmental Objectives and Build Competence (A6) 19
Activity A5 — Design Life-Cycle Operational Controls (A5) 21
Activity A6 — Run an Environmental Emergency Tabletop (A3, A4) 23
Activity A7 — Evaluate Performance and Conduct an Internal Audit (A5, A6) 25
Activity A8 — Close Corrective Action and Plan the 2026 Transition (A7, A8) 27
7. Assessment Preparation **29**
Glossary **29**
References and Further Reading **29**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

How to Use This Guide

This Learner Guide supports Fundamentals of ISO 14001 Environmental Management System (EMS) (TGS-2023037592). The presentation is concept-led; detailed working procedures are contained here and in the eight self-contained activity folders.

Standard and legal boundary
This guide paraphrases ISO 14001 concepts for learning and is not the copyrighted standard, legal advice or a substitute for competent regulatory interpretation. Use a licensed copy of ISO 14001:2026 for formal conformity work and verify Singapore obligations against current authoritative sources.

Current edition
As at 20 August 2026, ISO 14001:2026, Environmental management systems - Requirements with guidance for use. Edition 4 was published on 15 April 2026. ISO 14001:2015 and ISO 14001:2015/Amd 1:2024 are withdrawn; their climate considerations are incorporated into the current edition.

Course Learning Outcomes

- LO1: Identify environmental hazards and assess risks within the workplace context using an ISO 14001 EMS.
- LO2: Determine actions for environmental accidents and report unsafe conditions or acts to relevant authorities.
- LO3: Evaluate environmental hazards and oversee ISO 14001 EMS-related training and assessment activities.
- LO4: Enhance the intended outcomes of environmental management through ISO 14001 EMS continual improvement.

TSC Knowledge and Abilities

Code	Knowledge / ability
K1	Purpose, benefits, structure and PDCA logic of an ISO 14001 environmental management system
K2	Organisation context, interested parties, EMS scope, leadership, environmental policy and responsibilities
K3	Environmental hazards, aspects, impacts, significance, risks and opportunities, and compliance obligations
K4	Objectives, support, competence, communication, documented information, life-cycle perspective and operational control
K5	Monitoring, measurement, analysis, compliance evaluation, internal audit and management review
K6	Emergency response, nonconformity, corrective action, continual improvement and the ISO 14001:2026 changes
A1	Identify environmental hazards and environmental aspects from workplace evidence
A2	Assess environmental impacts, significance, risks

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

	and opportunities using defined criteria
A3	Determine proportionate actions for environmental incidents and emergency situations
A4	Escalate and report environmental unsafe conditions, acts and compliance concerns
A5	Evaluate environmental hazards, operational controls and environmental performance evidence
A6	Plan and oversee EMS competence, awareness, training and evaluation activities
A7	Identify and prioritise opportunities to improve EMS performance and intended outcomes
A8	Document corrective action, effectiveness review and continual improvement according to organisational procedures

Connected Case Organisation

Name: Meridian EcoPack Pte Ltd (MEP)

Profile: a fictional Singapore SME with 120 employees producing printed corrugated and fibre-based packaging for regional food, electronics and healthcare customers.

Customers: Two key customers require evidence of an ISO 14001:2026-aligned EMS and stronger value-chain environmental controls before contract renewal.

Symptoms: Electricity intensity rose 11%; mixed production waste increased; an ink tote spill nearly reached a drain; wastewater and contractor records are incomplete; nearby tenants reported intermittent odour; supplier environmental evidence varies.

Goal: Build an evidence-led EMS that protects the environment, meets applicable obligations, improves performance and supports certification readiness.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

1. EMS Foundations and ISO 14001

An environmental management system (EMS) is the connected set of policies, responsibilities, processes, resources, controls and evidence used to manage environmental aspects, meet compliance obligations and improve environmental performance. It should be integrated with normal management rather than operated as a separate certificate project.

Family document	Purpose in an EMS
ISO 14001	Auditable EMS requirements with guidance for use.
ISO 14004	General guidance for phased EMS establishment, implementation, maintenance and improvement.
ISO 14031	Guidance on environmental performance evaluation.
ISO 14040/14044	Principles, framework and requirements for life-cycle assessment.
ISO 19011	Guidance for management-system audit programmes and audits.

Core working terms

Term	Working distinction	MEP example
Environmental aspect	Element of an activity, product or service that interacts or can interact with the environment.	Ink storage, electricity use or paper offcuts.
Environmental impact	Change to the environment resulting wholly or partly from an aspect.	Water contamination, emissions or resource depletion.
Environmental condition	State or characteristic of the environment at a point in time.	Flood exposure, air quality, water stress or sensitive receptors.
Hazard and risk	A hazard is a potential source of harm; risk considers uncertainty and consequence. EMS planning also considers opportunities.	A spill pathway to a drain and the likelihood/consequence of pollution.
Compliance obligation	Legal requirement or other requirement an organisation has to or chooses to comply with.	Permit condition, applicable regulation or customer commitment.
Environmental performance	Measurable results related to management of environmental aspects.	kWh per tonne, wastewater result or waste-diversion rate.

PDCA and intended outcomes

PDCA	Management question	Typical EMS evidence
Plan	What context, aspects, obligations, risks, opportunities and objectives matter?	Context review, registers and plans.
Do	Are competent people applying life-cycle and operational controls?	Operating criteria, supplier controls, competence and emergency arrangements.
Check	What happened, are obligations fulfilled and is the EMS effective?	Monitoring, compliance evaluation, audit and management review.
Act	What must be corrected, improved or changed?	Corrective action, effectiveness review and improvement decisions.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Certification is third-party assurance of an organisation's defined EMS scope; ISO does not certify organisations. A typical journey includes leadership commitment, gap analysis, implementation and evidence accumulation, internal audit and management review, certification audit stages, corrective action and surveillance.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

2. Context, Leadership and Planning — Clauses 4 to 6

Clause area	Practical intent	Useful objective evidence
4.1 Context	Identify internal/external issues and environmental conditions that affect intended outcomes or are affected by the organisation.	Context scan, environmental-condition evidence, strategic review.
4.2 Interested parties	Determine relevant parties, needs and which needs become compliance obligations.	Party register, contracts, regulator/customer evidence.
4.3 Scope	Set defensible boundaries considering context, functions, sites, activities, products/services and life-cycle perspective.	Controlled scope and applicability rationale.
4.4 EMS	Operate and improve interacting EMS processes.	Process map, owners, inputs/outputs, measures and records.
5.1 Leadership	Integrate EMS requirements, provide resources and support people with relevant roles.	Decisions, budgets, review participation and authority.
5.2 Policy	Set direction and commitments appropriate to context.	Approved policy, availability and understanding.
5.3 Roles	Assign, communicate and support responsibilities and authorities.	Role matrix, incident authority and reporting routes.
6.1 Planning	Determine aspects, obligations, risks/opportunities and actions, including abnormal and emergency conditions.	Registers, criteria, plans and effectiveness review.
6.2 Objectives	Set measurable objectives and plan execution/evaluation.	Baselines, KPI definitions, targets, milestones and results.
6.3 Change	Plan and manage EMS-relevant changes in a controlled manner.	Impact review, safeguards, approval and verification.

A defensible aspect-significance method

1. Define the process, activity, product/service, boundary and operating condition.
2. Identify inputs, outputs, releases, resource use, waste and credible emergencies.
3. Write the aspect as the interaction/cause and the impact as the environmental change.
4. Apply documented criteria such as severity, likelihood, compliance exposure, stakeholder concern and control effectiveness.
5. Apply explicit override rules for legal, emergency or stakeholder triggers.
6. Separate inherent significance from residual risk after controls; record uncertainty and evidence.
7. Link priorities to controls, objectives, monitoring and improvement.
8. Review after incidents, change, new evidence, changing obligations or planned intervals.

Inside-out and outside-in
Inside-out analysis asks how MEP affects the environment. Outside-in analysis asks how environmental conditions such as climate, biodiversity, resource availability or flooding affect MEP's ability to achieve EMS outcomes. Both inform context and planning.

Compliance obligations workflow

9. Identify statutory, regulatory, permit, contract, customer and voluntary sources.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

10. Verify currency and applicability with authoritative sources and competent advice where needed.
11. Link a plain-language applicability statement to a process and aspect.
12. Define control, evidence, owner, frequency and escalation.
13. Periodically evaluate compliance and record evidence, status and evaluator.
14. Monitor changes and update controls, competence and information.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

3. Support and Operation — Clauses 7 and 8

Area	What effective implementation looks like	Evidence question
Resources	People, infrastructure, technology, finance and specialist support fit EMS risks and plans.	Are resources enough for controls and emergencies?
Competence	Criteria are role-specific; gaps are addressed; action effectiveness is evaluated.	Can the person perform correctly, not merely show attendance?
Awareness	People understand policy, significant aspects, contribution and consequences.	Can operators explain controls and escalation?
Communication	What, when, with whom, how and who are defined.	Are incident and external messages controlled and retained?
Documented information	Information is suitable, available, protected, controlled and retained.	Is the current instruction available and can records be retrieved?
Operational control	Operating criteria address relevant causes and value-chain stages.	Do controls show implementation and effectiveness?
External provision	Requirements and oversight are proportionate to control or influence.	Do suppliers and contractors meet defined criteria?
Emergency readiness	All potential emergencies are identified, prevented where possible, rehearsed and improved.	Are roles, equipment, receptors, recovery and learning covered?

Life-cycle perspective without requiring a full LCA

- Consider design/material selection, purchasing, operations, delivery/use where influenced and end-of-life.
- Distinguish direct control from influence through specifications, purchasing, contracts or guidance.
- Use proportionate environmental criteria and evidence for external providers.
- Avoid shifting impacts between stages—for example, lighter packaging that increases damage.

Emergency preparedness and response

15. Identify credible emergency situations and receptors/pathways.
16. Prevent or mitigate adverse impacts where practicable.
17. Define authority, roles, alarms, containment, communications, notification assessment, recovery and evidence protection.
18. Provide equipment and competent people; coordinate contractors/responders.
19. Test arrangements periodically where practicable and after change.
20. Evaluate tests or events; correct gaps and retain learning.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

4. Performance Evaluation — Clause 9

Monitoring generates observations. Measurement assigns values. Analysis identifies patterns. Evaluation compares evidence with criteria to support a decision. Keeping these distinct prevents a dashboard from being mistaken for a compliance or effectiveness conclusion.

Element	Design questions	Example evidence
Indicators	What, method, unit/boundary, frequency, owner, threshold and data-quality control?	kWh per tonne, wastewater result, diversion and control completion.
Compliance evaluation	Which obligation/evidence, by whom, when and with what status/action?	Evaluation record, source/version, sample and escalation.
Internal audit	What objectives, scope, criteria, methods, independence and risk-informed sampling?	Programme, plan, evidence log, findings and follow-up.
Management review	What changes, trends, objectives, compliance, audits, resources and opportunities require decisions?	Minutes with changes, resources, owners, dates and follow-up.

Writing a defensible audit finding

- Criterion: identify the applicable requirement or organisational control.
- Objective evidence: state verifiable facts, sample and references.
- Gap: explain the difference without prescribing a solution.
- Risk/context: describe credible consequence and scope.
- Classification: apply defined programme rules consistently.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

5. Improvement and ISO 14001:2026 Transition — Clause 10

Response	Purpose	MEP example
Correction	Control or correct the detected issue.	Replace obsolete contractor contacts.
Containment	Protect receptors, processes and evidence.	Isolate the drain, stop release and secure affected material.
Corrective action	Eliminate verified cause and reduce recurrence.	Redesign ownership/change controls for emergency readiness.
Effectiveness review	Test sustained intended results.	Sample shifts and repeat a timed drill against criteria.
Continual improvement	Improve EMS and environmental performance.	Use trend, audit, incident and review evidence.

A controlled 2026 transition

21. Confirm transition arrangements with the accredited certification body.
22. Perform a clause/process gap review using a licensed ISO 14001:2026 copy.
23. Prioritise environmental/compliance risk, dependencies and evidence maturity.
24. Address environmental conditions, outcomes, leadership support, planned change, value-chain oversight, all potential emergencies, audit objectives and management-review results.
25. Assign owners, resources, milestones and competence/document changes.
26. Implement without weakening existing controls; verify effectiveness.
27. Complete internal audit and management review before the transition audit.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

6. Detailed Activity Procedures

The slide deck provides only scenario, outputs and decision prompts. Use the matching workbook and checklist in each activity folder. Outputs form one connected MEP evidence chain.

Activity A1 — Build the Environmental Baseline and Aspect Inventory (A1)

Identify workplace environmental hazards and aspects from process and site evidence.

Scenario
MEP is preparing its first ISO 14001:2026-aligned EMS. Managers have energy bills and waste invoices, but no shared view of where activities interact with air, water, land, resources or neighbours.

Expected output: A process-boundary map, environmental baseline and aspect-impact inventory with evidence references.

Files: Scenario guide PDF, EMS Evidence Workbook (Excel), site-plan evidence cards

Decision questions

28. Which activities, products and services create direct or influenced environmental interactions?
29. For each interaction, what is the aspect and what is the resulting environmental impact?
30. What normal, abnormal and emergency conditions must be included, and what evidence is still missing?

Detailed step-by-step procedure

Step 1: Read the company profile, process flow, site plan and evidence cards before scoring anything.

How to complete and evidence: Read all evidence and record source, date and reviewer before deciding.

Step 2: Define the review boundary from receipt of paper and inks through printing, conversion, storage, dispatch and end-of-life influence.

How to complete and evidence: Complete the intended workbook fields in plain language; label assumptions and missing data.

Step 3: Walk each process stage and record inputs, outputs, releases, resource use, waste streams and potential emergencies.

How to complete and evidence: Apply the supplied criteria or decision rule consistently and retain a repeatable rationale.

Step 4: Write each aspect as a cause or interaction and each impact as a change to the environment.

How to complete and evidence: Make the decision explicit: control, priority, owner, date, threshold, escalation or follow-up.

Step 5: Tag the operating condition, control/influence level, evidence source, owner and missing data for every entry.

How to complete and evidence: Ask a peer to challenge one material assumption or unintended consequence and record the response.

Step 6: Peer-review for omissions at maintenance, contractors, utilities, transport and product end-of-life stages.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

How to complete and evidence: Reconcile the output with earlier activities and update linked evidence to prevent conflicting versions.

Acceptance test

The inventory covers normal, abnormal and potential emergency conditions; every aspect-impact pair is logically linked and traceable to supplied evidence or an explicit evidence gap.

- The workbook is complete, legible and uses the intended tabs.
- Every conclusion is traceable to evidence or labelled as an assumption/evidence gap.
- Owners, dates, measures, controls and next decisions are stated where applicable.
- The team can explain the PDCA link, ability code and environmental consequence.

Debrief and submission

- Which activities, products and services create direct or influenced environmental interactions?
- For each interaction, what is the aspect and what is the resulting environmental impact?
- What normal, abnormal and emergency conditions must be included, and what evidence is still missing?
- What evidence would most improve confidence?
- What will trigger review or escalation?

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activity A2 — Assess Significance, Risks and Opportunities (A2)

Assess environmental impacts and EMS risks/opportunities using transparent criteria.

Scenario
MEP's departments use different rating scales. Production prioritises cost, Facilities prioritises spill prevention, and Sales prioritises customer carbon requests. Leadership needs one defensible method.

Expected output: A significance matrix, prioritised aspects register and risk/opportunity action shortlist.

Files: Significance and Risk Workbook (Excel), scoring calibration checklist PDF

Decision questions

31. Which criteria make the decision repeatable without pretending that one score is scientific certainty?
32. Which legal or stakeholder triggers should override a low numerical score?
33. How do inside-out environmental impacts differ from outside-in environmental conditions affecting the business?

Detailed step-by-step procedure

Step 1: Agree definitions and anchors for severity, likelihood, compliance exposure, stakeholder concern and existing-control effectiveness.

How to complete and evidence: Read all evidence and record source, date and reviewer before deciding.

Step 2: Score independently before discussing to expose inconsistent assumptions.

How to complete and evidence: Complete the intended workbook fields in plain language; label assumptions and missing data.

Step 3: Apply the stated formula and mandatory-override rules; document the evidence behind each score.

How to complete and evidence: Apply the supplied criteria or decision rule consistently and retain a repeatable rationale.

Step 4: Rank significant aspects and separate inherent significance from residual risk after controls.

How to complete and evidence: Make the decision explicit: control, priority, owner, date, threshold, escalation or follow-up.

Step 5: Identify at least one risk and one opportunity linked to each priority aspect.

How to complete and evidence: Ask a peer to challenge one material assumption or unintended consequence and record the response.

Step 6: Run a sensitivity check by changing one uncertain score and record whether the priority decision changes.

How to complete and evidence: Reconcile the output with earlier activities and update linked evidence to prevent conflicting versions.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Acceptance test

Priority aspects are produced by the approved criteria and override rules; scores have evidence/rationale, uncertainties are visible and opportunity actions are not confused with positive impacts.

- The workbook is complete, legible and uses the intended tabs.
- Every conclusion is traceable to evidence or labelled as an assumption/evidence gap.
- Owners, dates, measures, controls and next decisions are stated where applicable.
- The team can explain the PDCA link, ability code and environmental consequence.

Debrief and submission

- Which criteria make the decision repeatable without pretending that one score is scientific certainty?
- Which legal or stakeholder triggers should override a low numerical score?
- How do inside-out environmental impacts differ from outside-in environmental conditions affecting the business?
- What evidence would most improve confidence?
- What will trigger review or escalation?

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activity A3 — Build the Compliance Obligations Register (A4)

Determine applicable obligations, ownership, evidence and escalation routes without making unsupported legal conclusions.

Scenario
MEP holds licences and service contracts in separate folders. A customer code, wastewater conditions, hazardous-substance storage controls and waste records are not linked to process owners or evidence.

Expected output: A compliance obligations register, applicability rationale, evidence map and escalation log.

Files: Compliance Register Workbook (Excel), obligation-screening checklist PDF, NEA source links

Decision questions

34. Which obligations are mandatory, which are voluntary commitments and which require specialist confirmation?
35. How does each obligation apply to a specific aspect, activity, site or contractor?
36. What evidence demonstrates current status and who must be told when a potential breach is found?

Detailed step-by-step procedure

Step 1: List each significant aspect and identify possible statutory, permit, contract, customer and voluntary sources.

How to complete and evidence: Read all evidence and record source, date and reviewer before deciding.

Step 2: Consult current authoritative sources and record the source, issue date, review date and responsible reviewer.

How to complete and evidence: Complete the intended workbook fields in plain language; label assumptions and missing data.

Step 3: Write a plain-language applicability statement and link the obligation to process, aspect and control.

How to complete and evidence: Apply the supplied criteria or decision rule consistently and retain a repeatable rationale.

Step 4: Define evidence, frequency, owner, evaluator and escalation route; mark any item requiring legal or technical advice.

How to complete and evidence: Make the decision explicit: control, priority, owner, date, threshold, escalation or follow-up.

Step 5: Test three sample records against the stated obligation and record compliant, noncompliant or insufficient-evidence status.

How to complete and evidence: Ask a peer to challenge one material assumption or unintended consequence and record the response.

Step 6: Schedule change monitoring and a periodic formal evaluation of compliance.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

How to complete and evidence: Reconcile the output with earlier activities and update linked evidence to prevent conflicting versions.

Acceptance test

Every register row has a source, applicability rationale, owner, evidence, review frequency and escalation route; unresolved interpretation is escalated rather than guessed.

- The workbook is complete, legible and uses the intended tabs.
- Every conclusion is traceable to evidence or labelled as an assumption/evidence gap.
- Owners, dates, measures, controls and next decisions are stated where applicable.
- The team can explain the PDCA link, ability code and environmental consequence.

Debrief and submission

- Which obligations are mandatory, which are voluntary commitments and which require specialist confirmation?
- How does each obligation apply to a specific aspect, activity, site or contractor?
- What evidence demonstrates current status and who must be told when a potential breach is found?
- What evidence would most improve confidence?
- What will trigger review or escalation?

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activity A4 — Set Environmental Objectives and Build Competence (A6)

Translate priority aspects into measurable objectives, action plans and role-based competence evidence.

Scenario
MEP's policy promises resource efficiency and pollution prevention, but targets have no baseline, owners or response thresholds. Annual awareness slides are delivered without checking competence.

Expected output: A SMART objective/KPI plan, action register, role-competence matrix and evaluation design.

Files: Objectives and Competence Workbook (Excel), objective-design and training-evaluation checklist PDF

Decision questions

37. Which indicators measure environmental performance, implementation progress and control effectiveness?
38. What resources, responsibilities, dates and evaluation methods make the objective executable?
39. What must each role be able to do, and what evidence shows competence rather than attendance?

Detailed step-by-step procedure

Step 1: Select two significant aspects and define the intended environmental result, baseline and data boundary.

How to complete and evidence: Read all evidence and record source, date and reviewer before deciding.

Step 2: Write measurable objectives with targets, dates, owners and assumptions.

How to complete and evidence: Complete the intended workbook fields in plain language; label assumptions and missing data.

Step 3: Define KPI formula, unit, source, frequency, data-quality control and action threshold.

How to complete and evidence: Apply the supplied criteria or decision rule consistently and retain a repeatable rationale.

Step 4: Build an action plan covering controls, resources, responsibilities, milestones and evaluation of results.

How to complete and evidence: Make the decision explicit: control, priority, owner, date, threshold, escalation or follow-up.

Step 5: Map critical EMS tasks to role-specific competence and awareness needs, including contractors.

How to complete and evidence: Ask a peer to challenge one material assumption or unintended consequence and record the response.

Step 6: Choose evaluation evidence such as observation, drill performance, work sample or knowledge demonstration and set review dates.

How to complete and evidence: Reconcile the output with earlier activities and update linked evidence to prevent conflicting versions.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Acceptance test

Each objective links to policy and a priority aspect; KPIs are reproducible and decision-ready; every critical role has competence criteria, evidence method, owner and review date.

- The workbook is complete, legible and uses the intended tabs.
- Every conclusion is traceable to evidence or labelled as an assumption/evidence gap.
- Owners, dates, measures, controls and next decisions are stated where applicable.
- The team can explain the PDCA link, ability code and environmental consequence.

Debrief and submission

- Which indicators measure environmental performance, implementation progress and control effectiveness?
- What resources, responsibilities, dates and evaluation methods make the objective executable?
- What must each role be able to do, and what evidence shows competence rather than attendance?
- What evidence would most improve confidence?
- What will trigger review or escalation?

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activity A5 — Design Life-Cycle Operational Controls (A5)

Evaluate significant aspects and design controls across activities and externally provided processes, products and services.

Scenario
MEP controls the printing line but has weak requirements for paper suppliers, cleaning contractors, waste collectors and transport providers. Instructions describe tasks but not environmental acceptance criteria.

Expected output: An operational control plan, supplier requirement matrix and control-verification schedule.

Files: Operational Control Workbook (Excel), life-cycle and supplier checklist PDF

Decision questions

40. Where can MEP control an activity directly and where can it only influence outcomes?
41. Which controls prevent or reduce impact at source before relying on administrative checks?
42. What acceptance evidence is proportionate for externally provided processes, products and services?

Detailed step-by-step procedure

Step 1: Map the value-chain stages from material selection through manufacture, delivery, use and end-of-life influence.

How to complete and evidence: Read all evidence and record source, date and reviewer before deciding.

Step 2: Link significant aspects and compliance obligations to each relevant stage and provider.

How to complete and evidence: Complete the intended workbook fields in plain language; label assumptions and missing data.

Step 3: Select controls using elimination/reduction at source, engineering, administrative and response layers.

How to complete and evidence: Apply the supplied criteria or decision rule consistently and retain a repeatable rationale.

Step 4: Define operating criteria, responsibility, evidence, monitoring method and change trigger for each control.

How to complete and evidence: Make the decision explicit: control, priority, owner, date, threshold, escalation or follow-up.

Step 5: Write supplier/contractor environmental requirements and proportionate approval or performance evidence.

How to complete and evidence: Ask a peer to challenge one material assumption or unintended consequence and record the response.

Step 6: Evaluate three current controls for implementation and effectiveness; record improvements and owners.

How to complete and evidence: Reconcile the output with earlier activities and update linked evidence to prevent conflicting versions.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Acceptance test

Controls address the actual cause of impact, include operating criteria and evidence, distinguish control from influence, and extend proportionately across relevant value-chain interfaces.

- The workbook is complete, legible and uses the intended tabs.
- Every conclusion is traceable to evidence or labelled as an assumption/evidence gap.
- Owners, dates, measures, controls and next decisions are stated where applicable.
- The team can explain the PDCA link, ability code and environmental consequence.

Debrief and submission

- Where can MEP control an activity directly and where can it only influence outcomes?
- Which controls prevent or reduce impact at source before relying on administrative checks?
- What acceptance evidence is proportionate for externally provided processes, products and services?
- What evidence would most improve confidence?
- What will trigger review or escalation?

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activity A6 — Run an Environmental Emergency Tabletop (A3, A4)

Determine, communicate and evaluate actions for a credible environmental emergency.

Scenario
At 10:20 a forklift punctures a 600-litre water-based ink tote. Product flows toward an internal drain. One operator raises the alarm; the quantity released, drain destination and regulatory notification threshold are initially uncertain.

Expected output: A time-stamped response log, notification decision record, containment plan and post-drill improvement actions.

Files: Emergency Tabletop Workbook (Excel), spill-response role cards and observer checklist PDF

Decision questions

43. What must happen in the first 5, 15 and 60 minutes, and who has authority to act?
44. Which conditions require internal escalation, external notification or specialist confirmation?
45. What evidence must be protected for recovery, investigation, compliance evaluation and learning?

Detailed step-by-step procedure

Step 1: Assign incident controller, safety, containment, communications, facilities and recorder roles.

How to complete and evidence: Read all evidence and record source, date and reviewer before deciding.

Step 2: Read the initial inject, stop unsafe work, raise the alarm and establish immediate human/environmental protection priorities.

How to complete and evidence: Complete the intended workbook fields in plain language; label assumptions and missing data.

Step 3: Identify the substance, quantity, pathway, receptors and control points; isolate the source only when safe.

How to complete and evidence: Apply the supplied criteria or decision rule consistently and retain a repeatable rationale.

Step 4: Deploy drain protection and spill containment conceptually using the site plan and available resources.

How to complete and evidence: Make the decision explicit: control, priority, owner, date, threshold, escalation or follow-up.

Step 5: Use the notification decision tree, record uncertainties, obtain competent advice and make time-stamped communications.

How to complete and evidence: Ask a peer to challenge one material assumption or unintended consequence and record the response.

Step 6: Plan waste recovery, clean-up verification, restoration and evidence retention.

How to complete and evidence: Reconcile the output with earlier activities and update linked evidence to prevent conflicting versions.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Step 7: Debrief against objectives; log corrective actions, owners, dates and the next drill or competence check.

How to complete and evidence: Check completion, traceability and ownership against the acceptance test before report-back.

Acceptance test

The team protects people and receptors, records decisions and uncertainties, uses the approved escalation path, preserves evidence and converts drill findings into owned improvement actions.

- The workbook is complete, legible and uses the intended tabs.
- Every conclusion is traceable to evidence or labelled as an assumption/evidence gap.
- Owners, dates, measures, controls and next decisions are stated where applicable.
- The team can explain the PDCA link, ability code and environmental consequence.

Debrief and submission

- What must happen in the first 5, 15 and 60 minutes, and who has authority to act?
- Which conditions require internal escalation, external notification or specialist confirmation?
- What evidence must be protected for recovery, investigation, compliance evaluation and learning?
- What evidence would most improve confidence?
- What will trigger review or escalation?

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activity A7 — Evaluate Performance and Conduct an Internal Audit (A5, A6)

Evaluate environmental performance, compliance evidence and EMS effectiveness through risk-based audit sampling.

Scenario
MEP reports that waste diversion improved, but production volume changed and two months of wastewater data are missing. The prior audit programme covered every department equally and did not state objectives.

Expected output: A KPI interpretation, compliance-evaluation record, audit plan, evidence log and defensible findings.

Files: Performance and Audit Workbook (Excel), audit role-play evidence pack and checklist PDF

Decision questions

46. What can the data support, what is uncertain and what requires normalisation or validation?
47. How is evaluation of compliance different from routine monitoring and from internal audit?
48. Does each finding state the criterion, objective evidence and gap without prescribing a solution?

Detailed step-by-step procedure

Step 1: Verify KPI definitions, boundaries, units, calibration/data-quality controls and missing-data treatment.

How to complete and evidence: Read all evidence and record source, date and reviewer before deciding.

Step 2: Normalise energy, water and waste measures to production where appropriate and interpret trends against thresholds.

How to complete and evidence: Complete the intended workbook fields in plain language; label assumptions and missing data.

Step 3: Evaluate a sample of obligations using current evidence and record status, basis, evaluator and action.

How to complete and evidence: Apply the supplied criteria or decision rule consistently and retain a repeatable rationale.

Step 4: Set audit objectives, scope, criteria, methods, timing and independence using risk and prior performance.

How to complete and evidence: Make the decision explicit: control, priority, owner, date, threshold, escalation or follow-up.

Step 5: Follow one aspect from policy and planning through operation, monitoring, competence and improvement evidence.

How to complete and evidence: Ask a peer to challenge one material assumption or unintended consequence and record the response.

Step 6: Triangulate interview, observation and document evidence; write findings and conduct a concise closing meeting.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

How to complete and evidence: Reconcile the output with earlier activities and update linked evidence to prevent conflicting versions.

Step 7: Prepare management-review inputs and decisions needed from the combined evidence.

How to complete and evidence: Check completion, traceability and ownership against the acceptance test before report-back.

Acceptance test

Conclusions respect data limitations; compliance status is evidence-based; the audit trail is reproducible; every finding contains criterion, evidence and gap; training follow-up is tied to observed competence needs.

- The workbook is complete, legible and uses the intended tabs.
- Every conclusion is traceable to evidence or labelled as an assumption/evidence gap.
- Owners, dates, measures, controls and next decisions are stated where applicable.
- The team can explain the PDCA link, ability code and environmental consequence.

Debrief and submission

- What can the data support, what is uncertain and what requires normalisation or validation?
- How is evaluation of compliance different from routine monitoring and from internal audit?
- Does each finding state the criterion, objective evidence and gap without prescribing a solution?
- What evidence would most improve confidence?
- What will trigger review or escalation?

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activity A8 — Close Corrective Action and Plan the 2026 Transition (A7, A8)

Document nonconformity, root cause, corrective action, effectiveness and controlled improvement.

Scenario
The tabletop showed that drain isolation ownership was unclear and contractor contact data was obsolete. Leadership also needs a structured gap review for the 2026 edition, including value-chain oversight and planned change.

Expected output: An NCR/CAPA record, effectiveness review, continual-improvement report and controlled transition plan.

Files: Corrective Action and Transition Workbook (Excel), report-quality checklist PDF

Decision questions

49. What is the immediate correction, what is the verified systemic cause and what similar risks exist elsewhere?
50. Which measures will demonstrate that the action remains effective after enough evidence has accumulated?
51. Which 2026 changes affect MEP's EMS, and how will the transition avoid disrupting current controls?

Detailed step-by-step procedure

Step 1: Record the nonconformity, requirement/criterion, objective evidence, affected scope and immediate correction.

How to complete and evidence: Read all evidence and record source, date and reviewer before deciding.

Step 2: Contain the risk and evaluate whether the issue exists in other shifts, drains, chemicals or contractor arrangements.

How to complete and evidence: Complete the intended workbook fields in plain language; label assumptions and missing data.

Step 3: Analyse cause using 5 Whys and barrier analysis; validate the selected root/system cause with evidence.

How to complete and evidence: Apply the supplied criteria or decision rule consistently and retain a repeatable rationale.

Step 4: Assign corrective actions to causes with owners, resources, dates and controlled-document/competence updates.

How to complete and evidence: Make the decision explicit: control, priority, owner, date, threshold, escalation or follow-up.

Step 5: Define effectiveness measures, sample size, review window and closure authority before implementation.

How to complete and evidence: Ask a peer to challenge one material assumption or unintended consequence and record the response.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Step 6: Complete the ISO 14001:2026 gap matrix covering context, leadership, risks/opportunities, planned change, value-chain controls, emergencies, audit objectives and management-review results.

How to complete and evidence: Reconcile the output with earlier activities and update linked evidence to prevent conflicting versions.

Step 7: Prioritise transition actions by environmental/compliance risk, dependency and readiness; present decisions required to management.

How to complete and evidence: Check completion, traceability and ownership against the acceptance test before report-back.

Step 8: Document results, residual risks, lessons learned and the next review in the continual-improvement report.

How to complete and evidence: Submit the workbook and explain evidence, decision, uncertainty and next review.

Acceptance test

Actions address verified causes, transition priorities are traceable to current requirements, effectiveness is measured after an appropriate interval, and a reviewer can reconstruct decisions, ownership, evidence and residual risk.

- The workbook is complete, legible and uses the intended tabs.
- Every conclusion is traceable to evidence or labelled as an assumption/evidence gap.
- Owners, dates, measures, controls and next decisions are stated where applicable.
- The team can explain the PDCA link, ability code and environmental consequence.

Debrief and submission

- What is the immediate correction, what is the verified systemic cause and what similar risks exist elsewhere?
- Which measures will demonstrate that the action remains effective after enough evidence has accumulated?
- Which 2026 changes affect MEP's EMS, and how will the transition avoid disrupting current controls?
- What evidence would most improve confidence?
- What will trigger review or escalation?

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

7. Assessment Preparation

The formal instruments assess the same knowledge and abilities developed in class. They are individual, open-book assessments; follow the current paper's complete candidate instructions.

Instrument	Duration	Coverage	Structure
Written Assessment (WA/SAQ)	60 minutes	K1–K6	Six open-ended questions; one per knowledge code.
Case Study (CS)	60 minutes	A1–A8	Four questions: A1–A2, A3–A4, A5–A6, A7–A8.

- Complete required attendance and identity fields before starting.
- Answer in your own words and connect decisions to scenario evidence.
- State criteria, evidence, conclusion, owner and next action.
- Submit both instruments and complete required assessment sign-off.

Glossary

Term	Working definition
Aspect	Element of activities, products or services that interacts or can interact with the environment.
Impact	Change to the environment resulting wholly or partly from environmental aspects.
Compliance obligation	Legal requirement or other requirement an organisation has to or chooses to comply with.
Environmental condition	State or characteristic of the environment at a point in time.
Life-cycle perspective	Consideration of relevant stages, proportionate to control or influence.
Nonconformity	Non-fulfilment of a requirement.
Correction	Action to eliminate a detected nonconformity.
Corrective action	Action to eliminate cause and prevent recurrence.
Objective evidence	Data supporting the existence or truth of something.
EMS	Environmental management system.
Effectiveness	Extent to which planned activities and results are achieved.
Environmental performance	Performance related to management of environmental aspects.

References and Further Reading

Legacy v7 trainer deck - content authority only: reference/WSQ - Master Trainer Slides - Fundamentals of ISO 14001 Environmental Management System (EMS) - v7.pptx

Course page - duration, outcomes, topics and assessment: https://www.tertiarycourses.com.sg/wsq-fundamentals-of-iso-14001-environmental-management-system-ems.html

ISO 14001:2026 standard status: https://www.iso.org/standard/14001

ISO 14001:2026 publication announcement: https://www.iso.org/news/2026/04/iso-14001-2026-published

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

ISO 14001:2026 changes explained: https://committee.iso.org/cms/live/live/en/sites/isoorg/contents/news/insights/Climate/iso-14001-2026-what%27s-changing.html

BSI ISO 14001:2026 key changes and guidance: https://groupsiteppd.bsigroup.com/en-US/products-and-services/standards-services/iso-14001-key-changes-and-guidance/

RRC certification, requirements and compliance guide: https://blog.rrc.co.uk/2026/07/29/a-guide-to-iso-14001-certification-preparation-requirements-staying-compliant/

TaxTMI ISO 14001 EMS overview: https://www.taxtmi.com/article/detailed?id=17161

Complaion ISO 14001 EMS guide: https://www.complaion.com/resources/guides/iso-14001-environmental-management-system

SCMEP ISO 14001 environmental management guide: https://www.scmep-online.org/iso-14001-environmental-management/

ImpactBuying ISO 14001 guide: https://impactbuying.com/iso-14001-2/

CTO Framework ISO 14001 requirements map: https://www.ctoframework.com/product/requirements/standards/ISO14001

Presencis clause 6.1 guide: https://presencis.com/regulations/iso-14001/article-6.1/

Sandland Packaging EMS implementation example: https://sandlandpackaging.co.uk/how-does-iso-14001-environmental-management-system-work/

CIVAC EMS implementation roadmap: https://civac.de/en/news/building-an-iso-14001-environmental-management-system

Leadity ISO 14001 requirements guide: https://leadity.de/en/unkategorisiert/iso-14001-requirements/

Singapore NEA environmental legislation portal: https://www.nea.gov.sg/corporate-functions/resources/legislation-international-law/legislation

Sources accessed for redevelopment on 20 August 2026. Official ISO and Singapore authority sources take precedence.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg